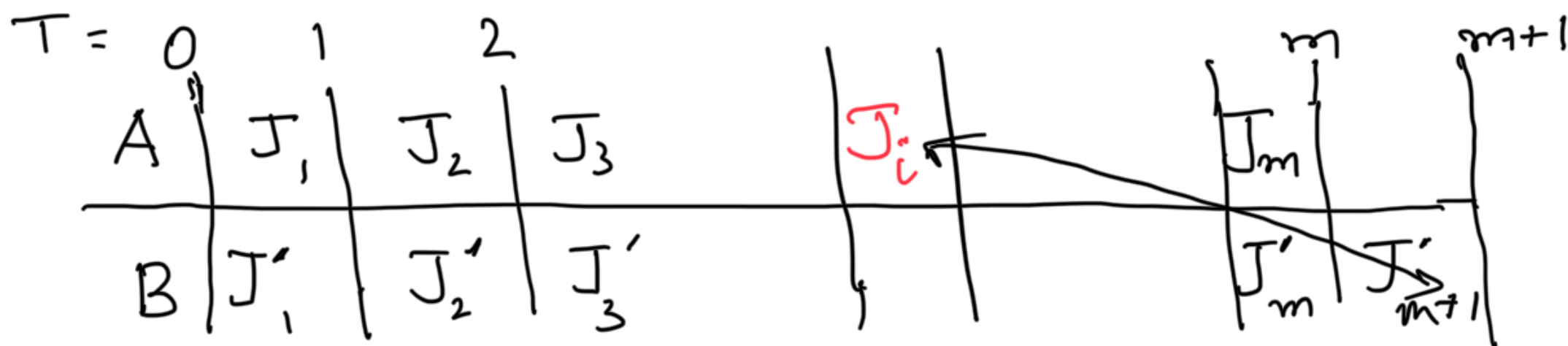


Lecture 17 DAA Mar 20

A: set of m jobs that can be scheduled

B: " " " " " " " "



Wlog consecutive time slots

Base Case

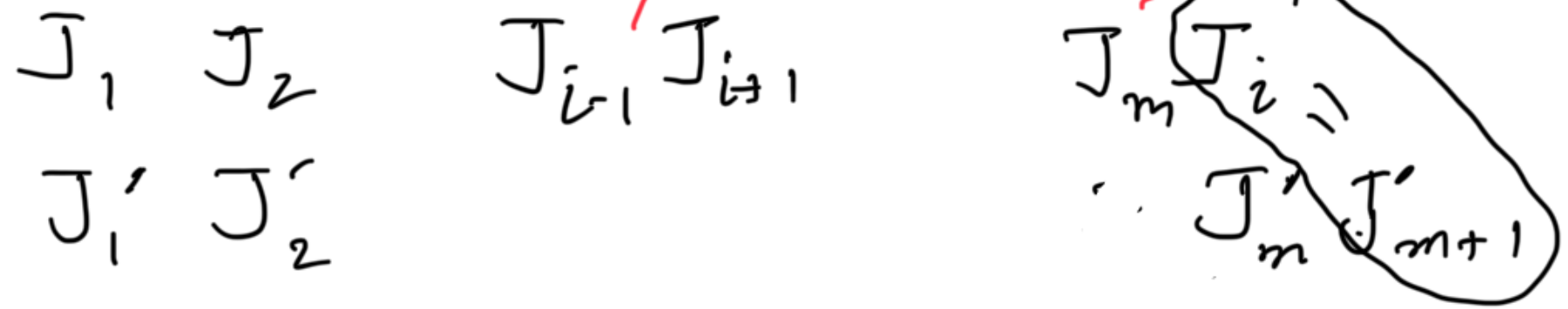
split case $m = 0$, i.e. A has no jobs, B has 1

Case 1: $J'_{m+1} \notin A$: just add J'_{m+1} to A

Otherwise

Suppose

$$J'_{m+1} = J_i$$

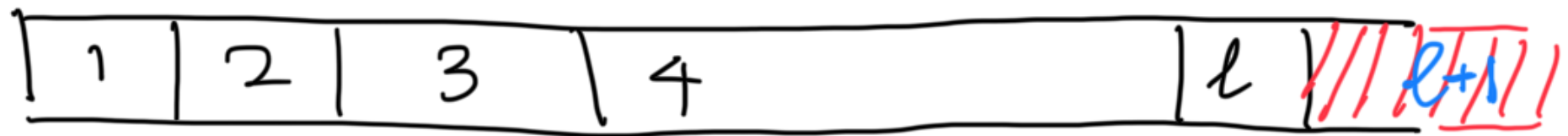


We can repeat the argument until Case 1 or base case, Exchange Property holds

Knap sack (revisited) $B=10$ Capacity

	1	2	3	4	5
Profit	10	4	32	27	28
weights	2	1	9	9	10
ratio	5	4	3.5	3	2.8

	1	2	3	4	...	l	$l+1$	...	n
profit	p_1	p_2	p_3			p_l	p_{l+1}		p_n
weights	w_1	w_2	w_3			w_l	w_{l+1}		w_n
decreasing ratio	$\frac{p_1}{w_1} \geq \frac{p_2}{w_2} \geq$					$\frac{p_l}{w_l} \geq \frac{p_{l+1}}{w_{l+1}} \geq$			$\frac{p_n}{w_n}$



max $\{ \overset{\text{profits}}{\text{the first } l \text{ objects}}, p_{l+1} \}$

the optimal value of the fractional knapsack is always $\geq$ 0-1 Knapsack

$$\sum_{j=1}^{l+1} p_j \geq \text{soln of fractional knapsack} \\ \geq \text{soln of 0-1 knapsack}$$

$$\max \left\{ \sum_{j=1}^l p_j, p_{l+1} \right\} \geq \frac{1}{2} \sum_{j=1}^{l+1} p_j \\ \geq \frac{1}{2} \text{ soln of 0-1 knapsack from above}$$

Heuristic that guarantees 50% of the optimal soln

(better than no guarantees)

Approximation algorithms that
run fast and provide guarantees on the
approx. optimality

n objects with profits P_i , weights w_i
 $1 \leq i \leq n$, B : capacity

$K(i, j)$: the optimal solution for
 knapsack for objects
 $\{1, 2, \dots, i\}$ and capacity j
 $1 \leq i \leq n$ $1 \leq j \leq B$

The original problem is $i = n$ $j = B$

$K(i, j)$ — if the i^{th} object is included
 $\swarrow$ max
 $P_i + K(i-1, j - w_i)$
 $\searrow$
 $K(i-1, j)$ if i^{th} object not included
 if < 0 , then $-\infty$

$K(i, i)$ —

$$K(i, j) = \max \{ P_i + K(i-1, j - w_i), K(i-1, j) \}$$

— Base case: row $i=1$, P_i if $w_i \leq j$, 0 otherwise

Time to fill up the $n \times B$ size table
 $= O(nB)$

The optimal value of the knapsack problem
 is available in $K(n, B)$

Is $O(nB)$ efficient?

Hint: $O(n \cdot 2^{\log B})$